

Muneeb Rehman, CT-25092
Lab 4

Question 1

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-4\CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [Debugger: TDM-GCC 9.2.0 64-bit Debug]
[Project: CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp]
1 // Exercise 1: Employee class with changeable name but constant ID
2
3 #include <iostream>
4 #include <string.h>
5
6 class Employee
7 {
8     private:
9         char* EmployeeName;
10        const int EmployeeId;
11
12    public:
13        Employee(const char* name, int id) : EmployeeId(id)
14        {
15            EmployeeName = new char[strlen(name) + 1];
16            strcpy(EmployeeName, name);
17        }
18
19        ~Employee()
20        {
21            delete[] EmployeeName;
22        }
23
24        const char* getEmployeeName()
25        {
26            return EmployeeName;
27        }
28
29        int getEmployeeId()
30        {
31            return EmployeeId;
32        }
33
34        void setEmployeeName(const char* name)
35        {
36            delete[] EmployeeName;
37            EmployeeName = new char[strlen(name) + 1];
38            strcpy(EmployeeName, name);
39        }
40    };
41
42
43 int main()
44 {
45     Employee Employee1("Ali Ahmed", 101);
46     Employee Employee2("Sara Khan", 102);
47     Employee Employee3("Hassan Raza", 103);
48
49     std::cout << "--- Initial Employee Details ---\n";
50     std::cout << "Employee 1: " << Employee1.getEmployeeName() << " (ID: " << Employee1.getEmployeeId() << ")\n";
51     std::cout << "Employee 2: " << Employee2.getEmployeeName() << " (ID: " << Employee2.getEmployeeId() << ")\n";
52     std::cout << "Employee 3: " << Employee3.getEmployeeName() << " (ID: " << Employee3.getEmployeeId() << ")\n";
53
54     Employee1.setEmployeeName("Ali Akbar");
55     Employee2.setEmployeeName("Sara Malik");
56
57     std::cout << "\n--- After Name Changes ---\n";
58     std::cout << "Employee 1: " << Employee1.getEmployeeName() << " (ID: " << Employee1.getEmployeeId() << ")\n";
59     std::cout << "Employee 2: " << Employee2.getEmployeeName() << " (ID: " << Employee2.getEmployeeId() << ")\n";
60     std::cout << "Employee 3: " << Employee3.getEmployeeName() << " (ID: " << Employee3.getEmployeeId() << ")\n";
61
62     return 0;
63 }
```

```
E:\University\University\SEME
--- Initial Employee Details ---
Employee 1: Ali Ahmed (ID: 101)
Employee 2: Sara Khan (ID: 102)
Employee 3: Hassan Raza (ID: 103)

--- After Name Changes ---
Employee 1: Ali Akbar (ID: 101)
Employee 2: Sara Malik (ID: 102)
Employee 3: Hassan Raza (ID: 103)

-----
Process exited after 0.1902 seconds with return value 0
Press any key to continue . . . |
```

Question 2

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-4\CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [Compiler: TDM-GCC 9.2.0 64-bit Debug]
[Project] Classes > CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp

1 // Exercise 2: DynamicArray class with dynamic memory allocation
2
3 #include <iostream>
4
5 class DynamicArray
6 {
7     private:
8         int* array;
9         int arraySize;
10        int currentIndex;
11
12    public:
13        DynamicArray(int size)
14        {
15            arraySize = size;
16            currentIndex = 0;
17            array = new int[arraySize];
18
19            for (int i = 0; i < arraySize; i++)
20            {
21                array[i] = 0;
22            }
23        }
24
25        ~DynamicArray()
26        {
27            delete[] array;
28        }
29
30        void add(int value)
31        {
32            if (currentIndex < arraySize)
33            {
34                array[currentIndex] = value;
35                currentIndex++;
36                std::cout << "Value " << value << " added successfully.\n";
37            }
38            else
39            {
40                std::cout << "Error: Array is full!\n";
41            }
42        }
43
44        int size()
45        {
46            return arraySize;
47        }
48
49        void display()
50        {
51        }
```

```
E:\University\University\SEMESTER-2\OOP\LAB\LAB-4\CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [Compiler: TDM-GCC 9.2.0 64-bit Debug]
[Project] Classes > CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp

51        std::cout << "Array contents: ";
52        for (int i = 0; i < arraySize; i++)
53        {
54            std::cout << array[i] << " ";
55        }
56        std::cout << "\n";
57    }
58};
59
60 int main()
61 {
62     int inputSize;
63
64     std::cout << "Enter size of array: ";
65     std::cin >> inputSize;
66
67     DynamicArray myArray(inputSize);
68
69     std::cout << "Array size: " << myArray.size() << "\n";
70
71     myArray.add(10);
72     myArray.add(20);
73     myArray.add(30);
74     myArray.add(40);
75
76     myArray.display();
77
78     return 0;
79 }
```

```
E:\University\University\SEME x + v
Enter size of array: 5
Array size: 5
Value 10 added successfully.
Value 20 added successfully.
Value 30 added successfully.
Value 40 added successfully.
Array contents: 10 20 30 40 0

-----
Process exited after 2.769 seconds with return value 0
Press any key to continue . . . |
```

Question 3

```
E:\University\University\SEMESTER 2\OOP\LAB\LAB-4\CT-25092_MuneebRehman - OOP_LAB_Assignment4.3.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
[Icons] [globals] TDM-GCC 9.2.0 64-bit Debug
Project Classes > CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.3.cpp
3 #include <iostream>
4
5 class Account
6 {
7 private:
8     int account_no;
9     float account_bal;
10    int security_code;
11    static int objectCount;
12
13 public:
14     Account()
15     {
16         objectCount++;
17     }
18
19     void initialize(int acc_no, float bal, int sec_code)
20     {
21         account_no = acc_no;
22         account_bal = bal;
23         security_code = sec_code;
24     }
25
26     void printData()
27     {
28         std::cout << "\n--- Account Details ---";
29         std::cout << "\nAccount Number: " << account_no;
30         std::cout << "\nAccount Balance: " << account_bal;
31         std::cout << "\nSecurity Code: " << security_code;
32         std::cout << "\n";
33     }
34
35     static int getObjectCount()
36     {
37         return objectCount;
38     }
39 };
40
41 int Account::objectCount = 0;
42
43 int main()
44 {
45     Account acc1;
46     acc1.initialize(12345, 50000.50, 1111);
47
48     Account acc2;
49     acc2.initialize(67890, 75000.75, 2222);
50
51     Account acc3;
52     acc3.initialize(11223, 100000.00, 3333);
53
54     acc1.printData();
55     acc2.printData();
56     acc3.printData();
57
58     std::cout << "\nTotal number of Account objects created: " << Account::getObjectCount() << "\n";
59
60     return 0;
61 }
```

```
E:\University\University\SEME X + -
--- Account Details ---
Account Number: 12345
Account Balance: 50000.5
Security Code: 1111

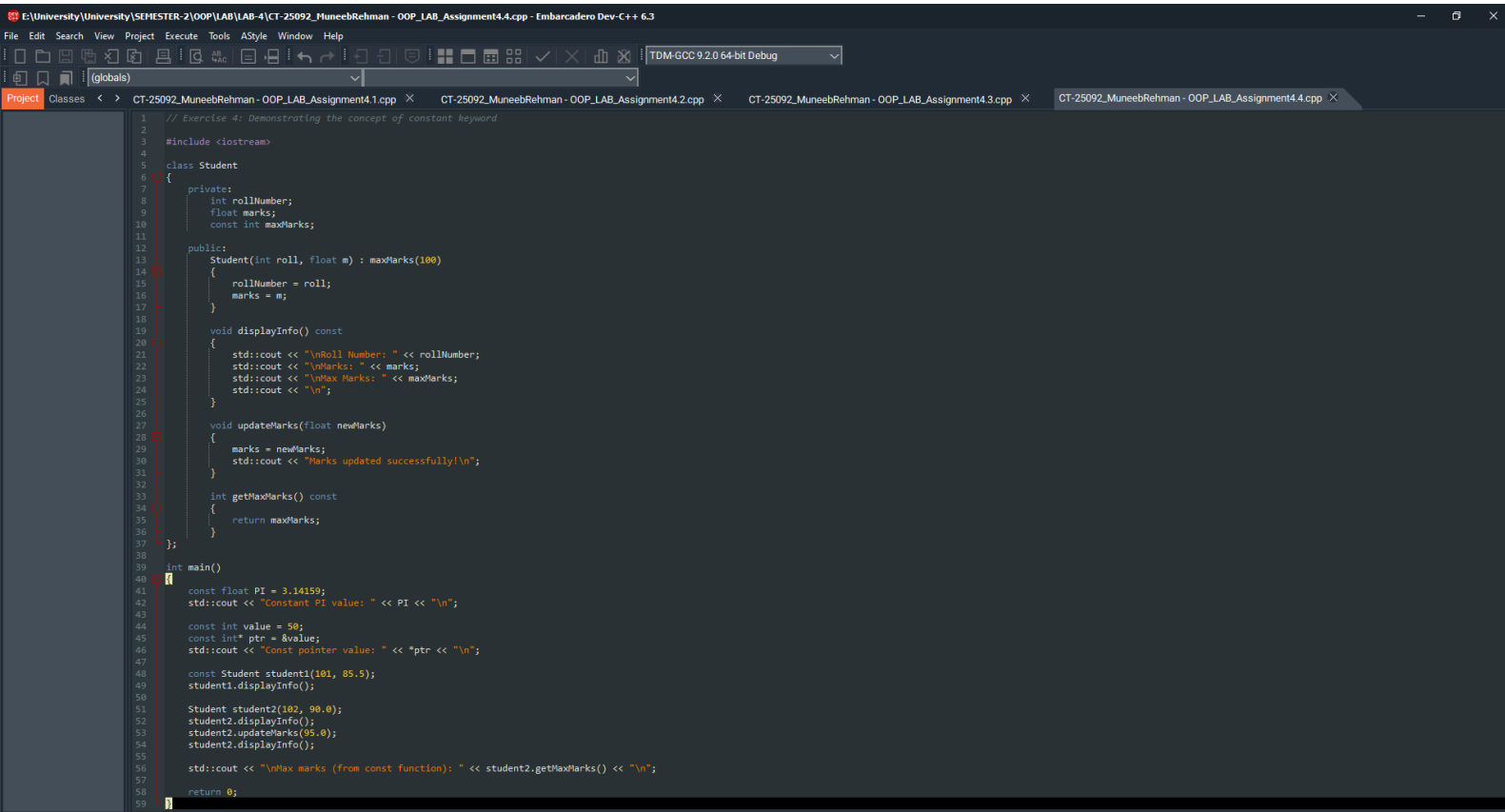
--- Account Details ---
Account Number: 67890
Account Balance: 75000.8
Security Code: 2222

--- Account Details ---
Account Number: 11223
Account Balance: 100000
Security Code: 3333

Total number of Account objects created: 3

-----
Process exited after 0.1804 seconds with return value 0
Press any key to continue . . . |
```

Question 4



```
1 // Exercise 4: Demonstrating the concept of constant keyword
2
3 #include <iostream>
4
5 class Student
6 {
7     private:
8         int rollNumber;
9         float marks;
10        const int maxMarks;
11
12    public:
13        Student(int roll, float m) : maxMarks(100)
14        {
15            rollNumber = roll;
16            marks = m;
17        }
18
19        void displayInfo() const
20        {
21            std::cout << "\nRoll Number: " << rollNumber;
22            std::cout << "\nMarks: " << marks;
23            std::cout << "\nMax Marks: " << maxMarks;
24            std::cout << "\n";
25        }
26
27        void updateMarks(float newMarks)
28        {
29            marks = newMarks;
30            std::cout << "Marks updated successfully!\n";
31        }
32
33        int getMaxMarks() const
34        {
35            return maxMarks;
36        }
37    };
38
39 int main()
40 {
41     const float PI = 3.14159;
42     std::cout << "Constant PI value: " << PI << "\n";
43
44     const int value = 50;
45     const int* ptr = &value;
46     std::cout << "Const pointer value: " << *ptr << "\n";
47
48     const Student student1(101, 85.5);
49     student1.displayInfo();
50
51     Student student2(102, 90.0);
52     student2.displayInfo();
53     student2.updateMarks(95.0);
54     student2.displayInfo();
55
56     std::cout << "\nMax marks (from const function): " << student2.getMaxMarks() << "\n";
57
58     return 0;
59 }
```

```
E:\University\University\SEME x + v
Constant PI value: 3.14159
Const pointer value: 50

Roll Number: 101
Marks: 85.5
Max Marks: 100

Roll Number: 102
Marks: 90
Max Marks: 100
Marks updated successfully!

Roll Number: 102
Marks: 95
Max Marks: 100

Max marks (from const function): 100

-----
Process exited after 0.1815 seconds with return value 0
Press any key to continue . . . |
```

Question 5

E:\University\University\SEHSTER-2\OOP\LAB\LAB-4\CT-25092_MuneebRehman - OOP_LAB_Assignment4.5.cpp - Embarcadero Dev-C++ 6.3

File Edit Search View Project Execute Tools AStyle Window Help

gdb (globals)

Project Classes < > CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.4.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment4.5.cpp

```
1 // Exercise 5: Hotel Mercato rent calculation system
2
3 #include <iostream>
4 #include <string.h>
5
6 class HotelRent
7 {
8 private:
9     char* customerName;
10    int numberOfDays;
11    const float dailyRate;
12    float totalRent;
13
14    void calculateRent()
15    {
16        if (numberOfDays > 7)
17        {
18            totalRent = (numberOfDays - 1) * dailyRate;
19        }
20        else
21        {
22            totalRent = numberOfDays * dailyRate;
23        }
24    }
25
26 public:
27    HotelRent() : dailyRate(1000.85)
28    {
29        customerName = NULL;
30        numberOfDays = 0;
31        totalRent = 0.0;
32    }
33
34    ~HotelRent()
35    {
36        if (customerName != NULL)
37        {
38            delete[] customerName;
39        }
40    }
41
42    void inputDetails()
43    {
44        char tempName[100];
45
46        std::cout << "Enter customer name: ";
47        std::cin.getline(tempName, 100);
48
49        customerName = new char[strlen(tempName) + 1];
50        strcpy(customerName, tempName);
51
52        std::cout << "Enter number of days stayed: ";
53        std::cin >> numberOfDays;
54
55        calculateRent();
56    }
57
58    void displayDetails() const
```

The screenshot shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment4.1.cpp". The code defines a "HotelRent" class with the following methods:

- `HotelRent()`: Constructor that initializes `customerName` to `NULL`, `numberOfDays` to `0`, and `totalRent` to `0.0`.
- `~HotelRent()`: Destructor that deletes `customerName` if it is not `NULL`.
- `void inputDetails()`: Method that prompts the user for customer name and number of days stayed, calculates the total rent, and displays the details.
- `void displayDetails() const`: Method that displays the hotel bill, including customer name, number of days, total rent, and a discount if the stay is more than 7 days.

The `main` function creates a `HotelRent` object, calls `inputDetails()`, `displayDetails()`, and returns `0`.


```
E:\University\University\SEME x + v
=== Hotel Mercato Rent Calculation System ===

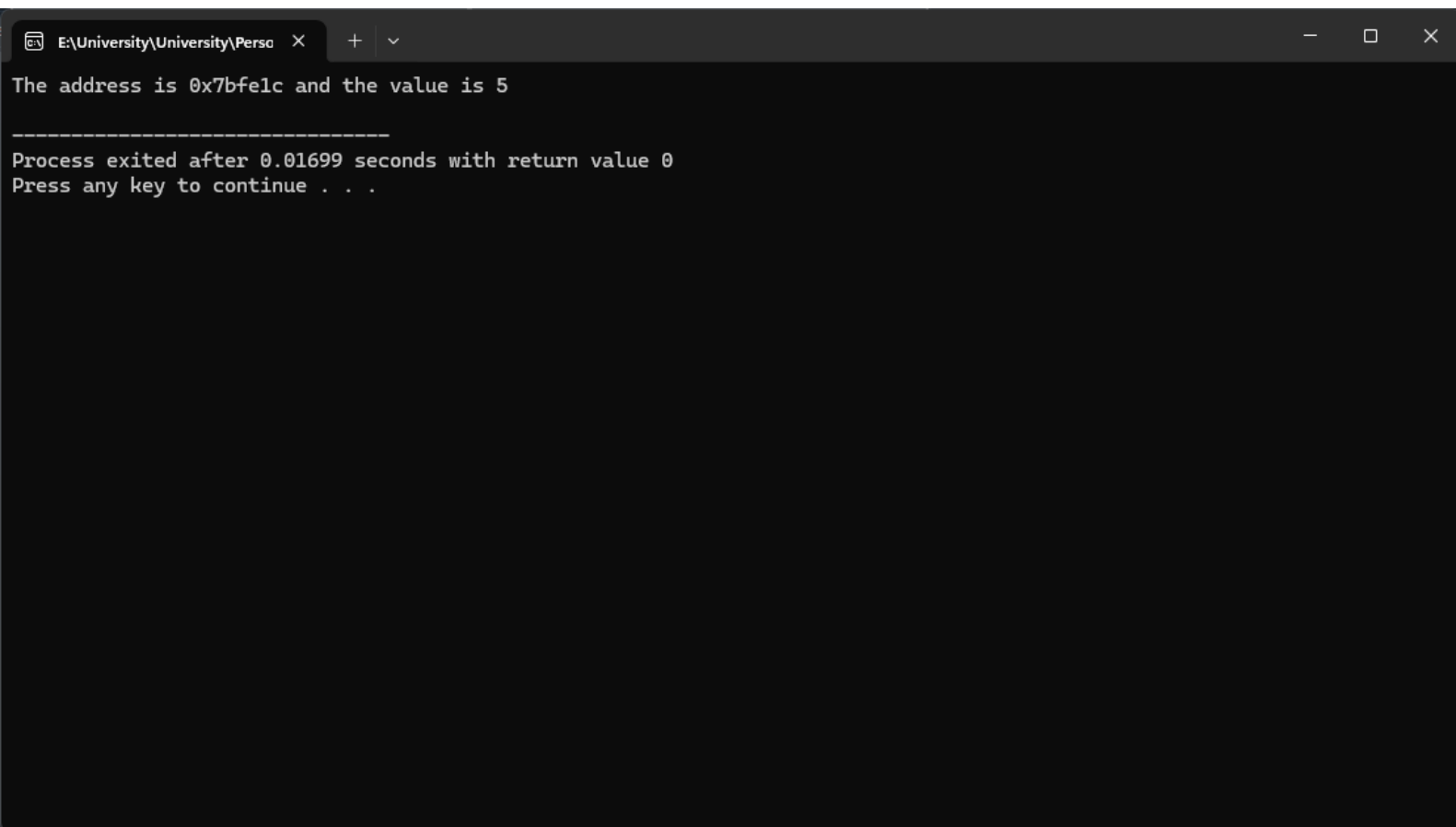
Enter customer name: Muneeb
Enter number of days stayed: 6

--- Hotel Mercato Bill ---
Customer Name: Muneeb
Days: 6
Rent: 6005.1 PKR

-----
Process exited after 3.953 seconds with return value 0
Press any key to continue . . . |
```

EXAMPLES

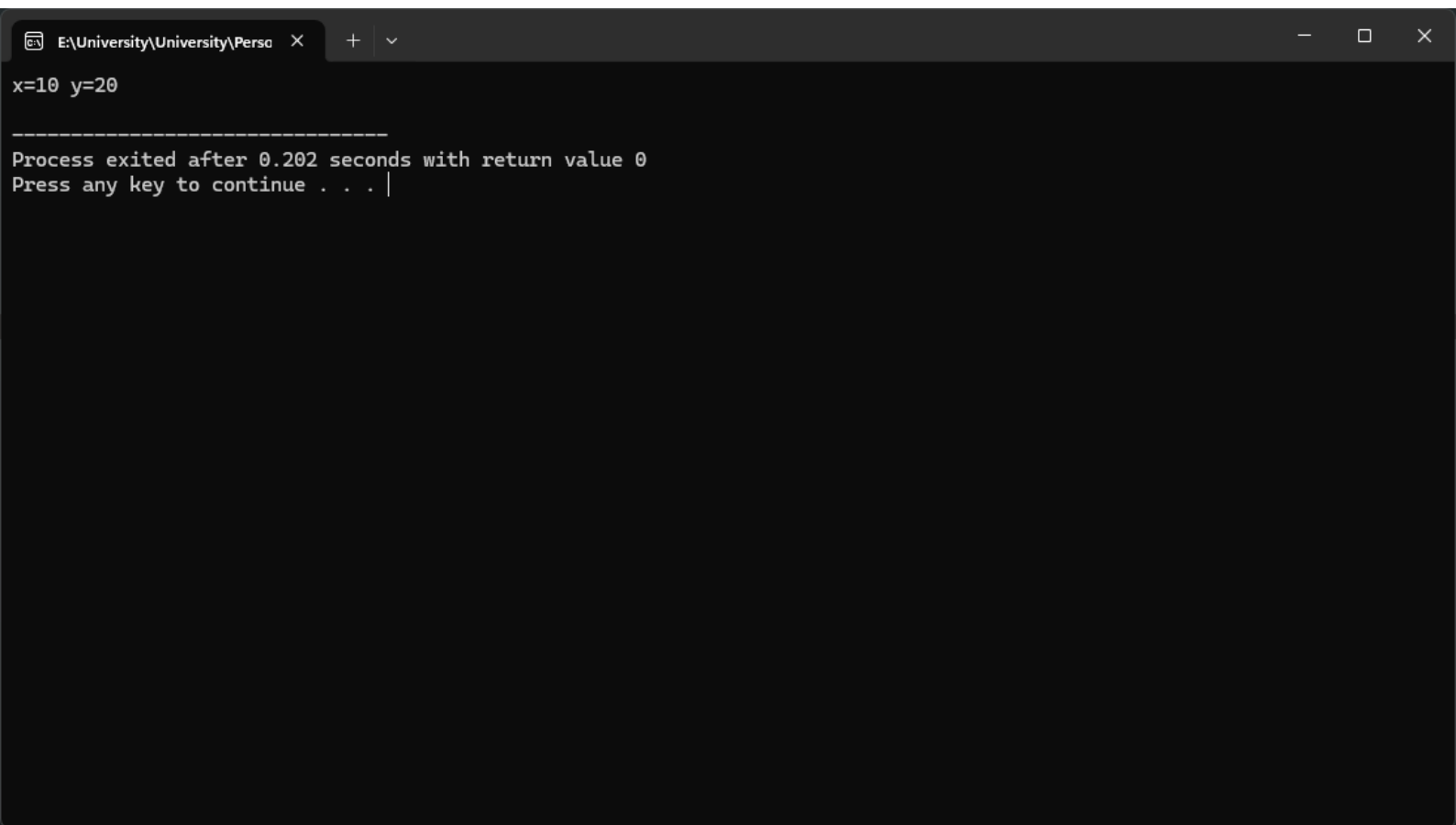
Example 1



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'E:\University\University\Perso' and includes standard window controls (minimize, maximize, close). The command prompt displays the following text:

```
The address is 0x7bfe1c and the value is 5  
  
-----  
Process exited after 0.01699 seconds with return value 0  
Press any key to continue . . .
```

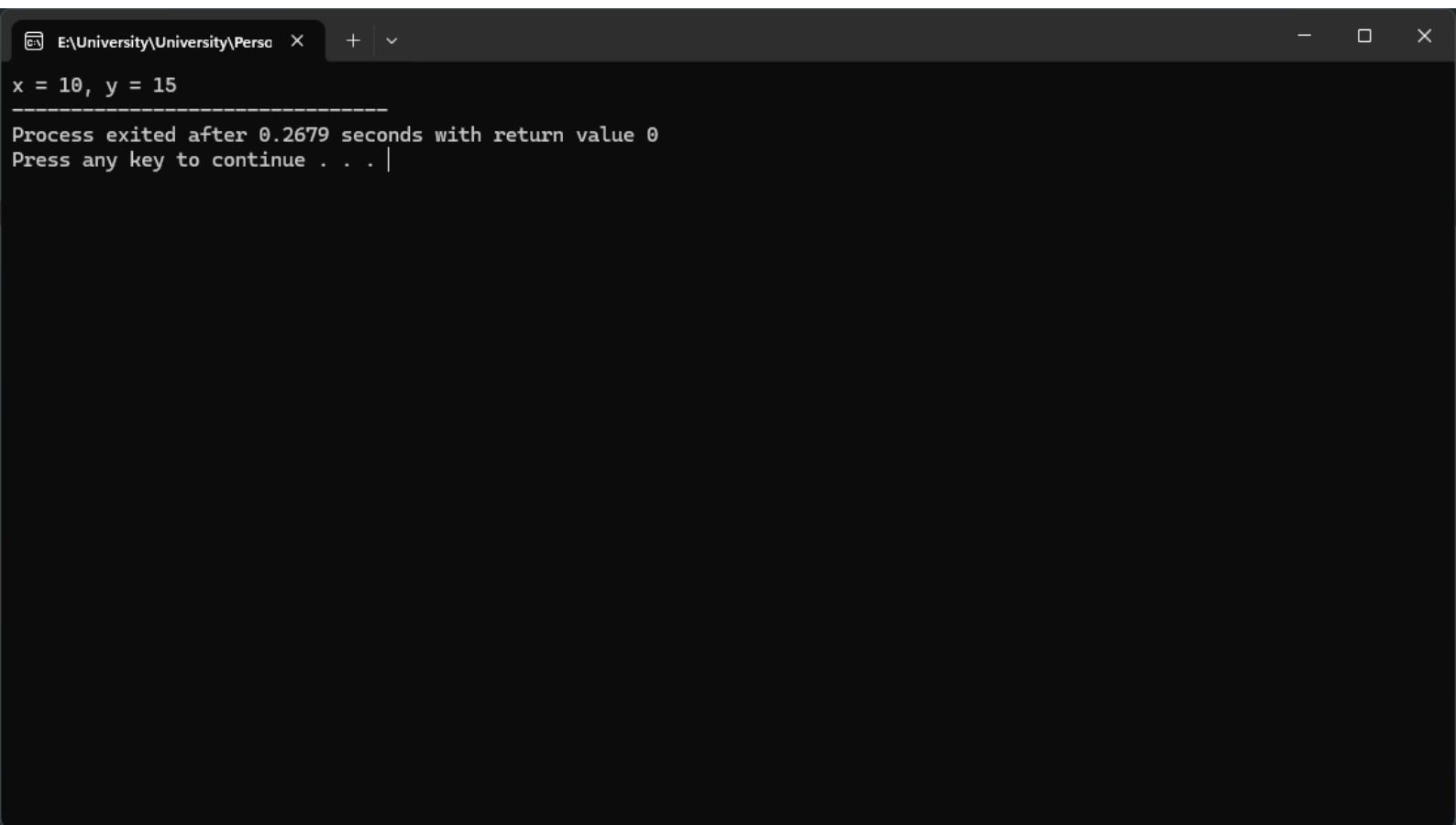
Example 2



A screenshot of a Windows command prompt window. The title bar shows the file path "E:\University\University\Perso" and standard window controls. The command prompt displays the output of a program: "x=10 y=20" followed by a separator line "-----". Below this, it states "Process exited after 0.202 seconds with return value 0" and "Press any key to continue . . . |".

```
E:\University\University\Perso x + v
x=10 y=20
-----
Process exited after 0.202 seconds with return value 0
Press any key to continue . . . |
```

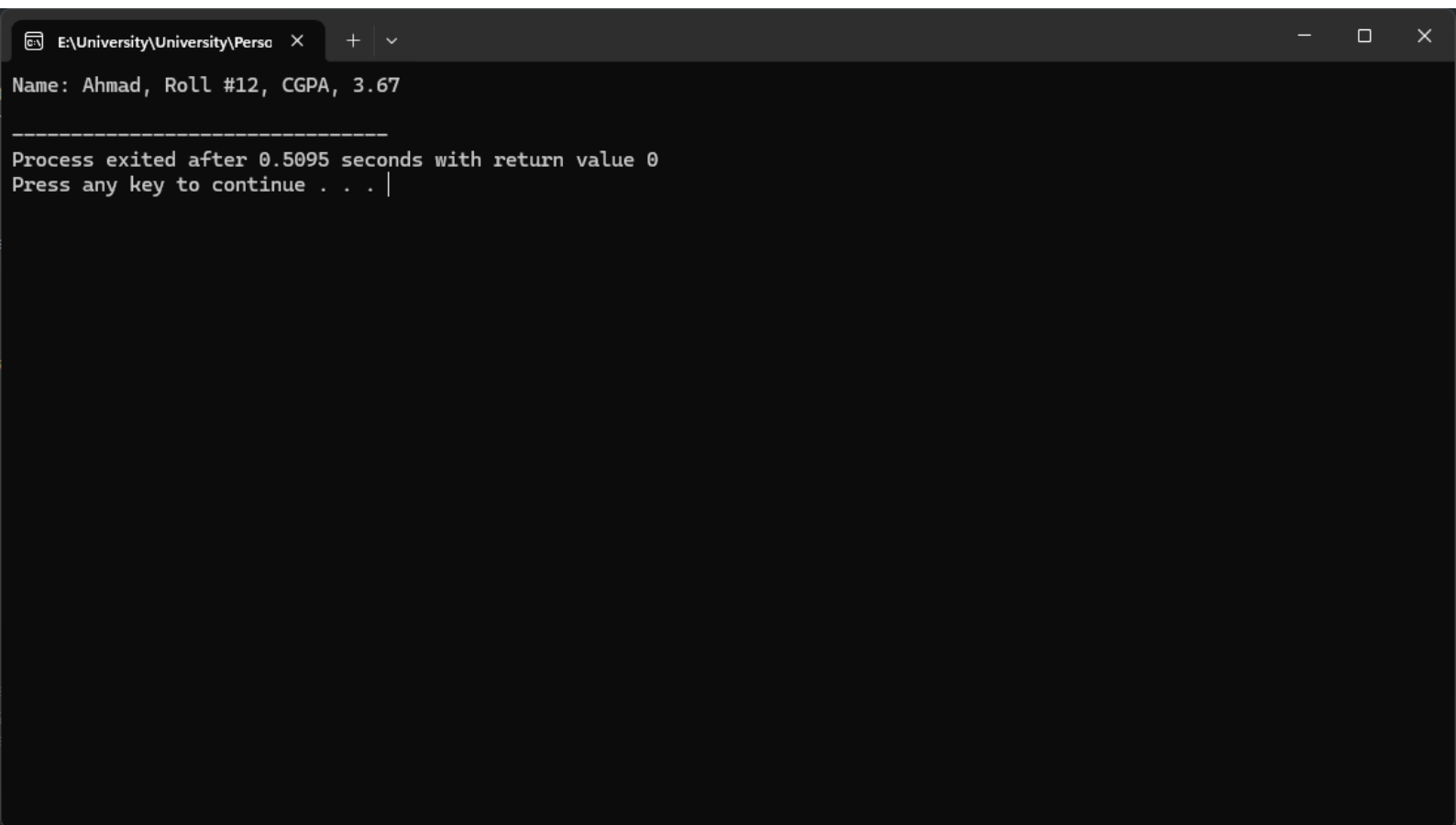
Example 3



A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\University\University\Perso' and standard window controls. The command prompt displays the output of a program: 'x = 10, y = 15' followed by a separator line, then 'Process exited after 0.2679 seconds with return value 0', and finally 'Press any key to continue . . . |'.

```
E:\University\University\Perso x + v
x = 10, y = 15
-----
Process exited after 0.2679 seconds with return value 0
Press any key to continue . . . |
```

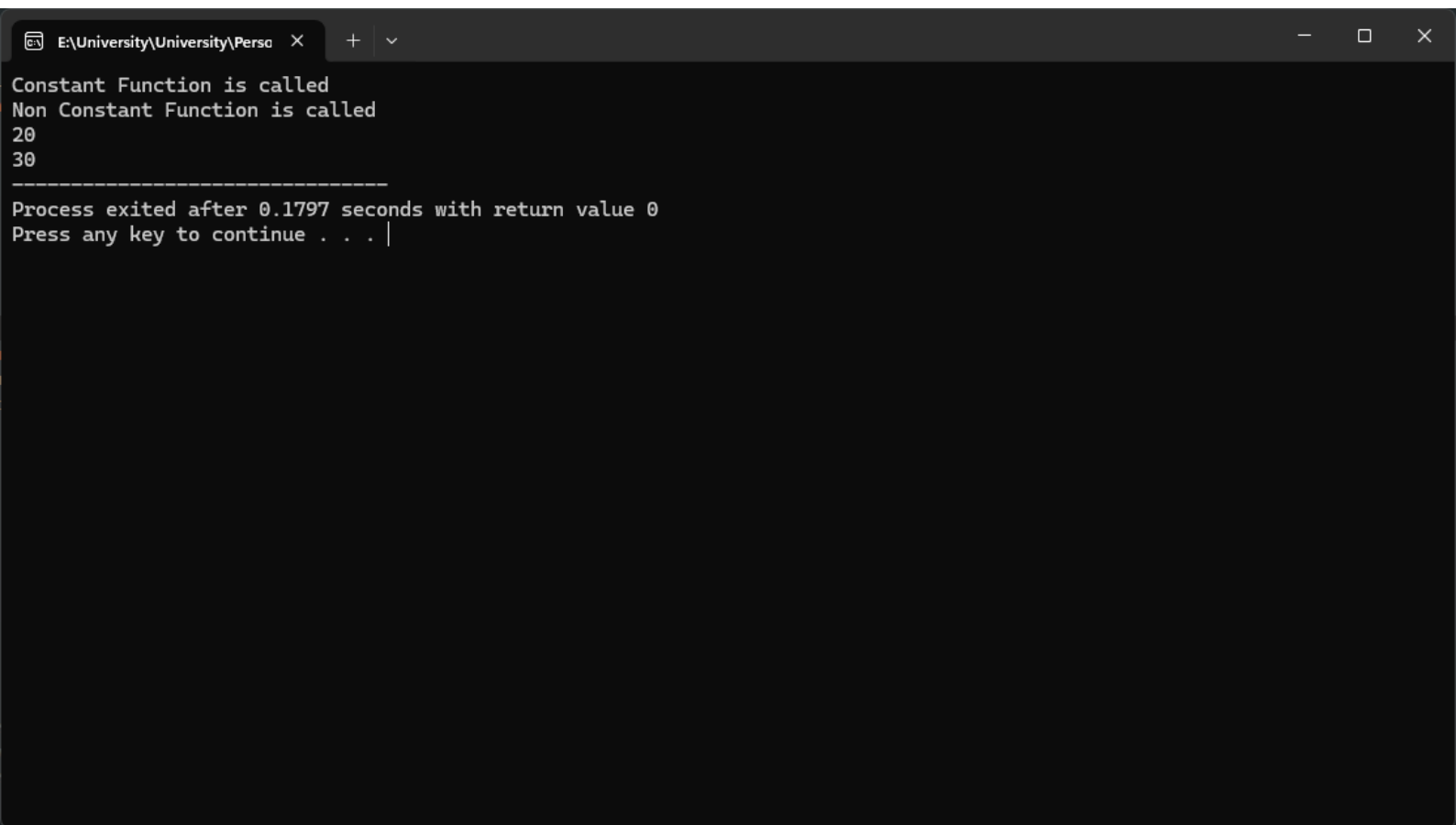
Example 4



A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\University\University\Perso' and standard window controls. The command prompt displays the output of a program: 'Name: Ahmad, Roll #12, CGPA, 3.67' followed by a separator line '-----'. Below this, it shows 'Process exited after 0.5095 seconds with return value 0' and 'Press any key to continue . . . |'.

```
E:\University\University\Perso >
Name: Ahmad, Roll #12, CGPA, 3.67
-----
Process exited after 0.5095 seconds with return value 0
Press any key to continue . . . |
```

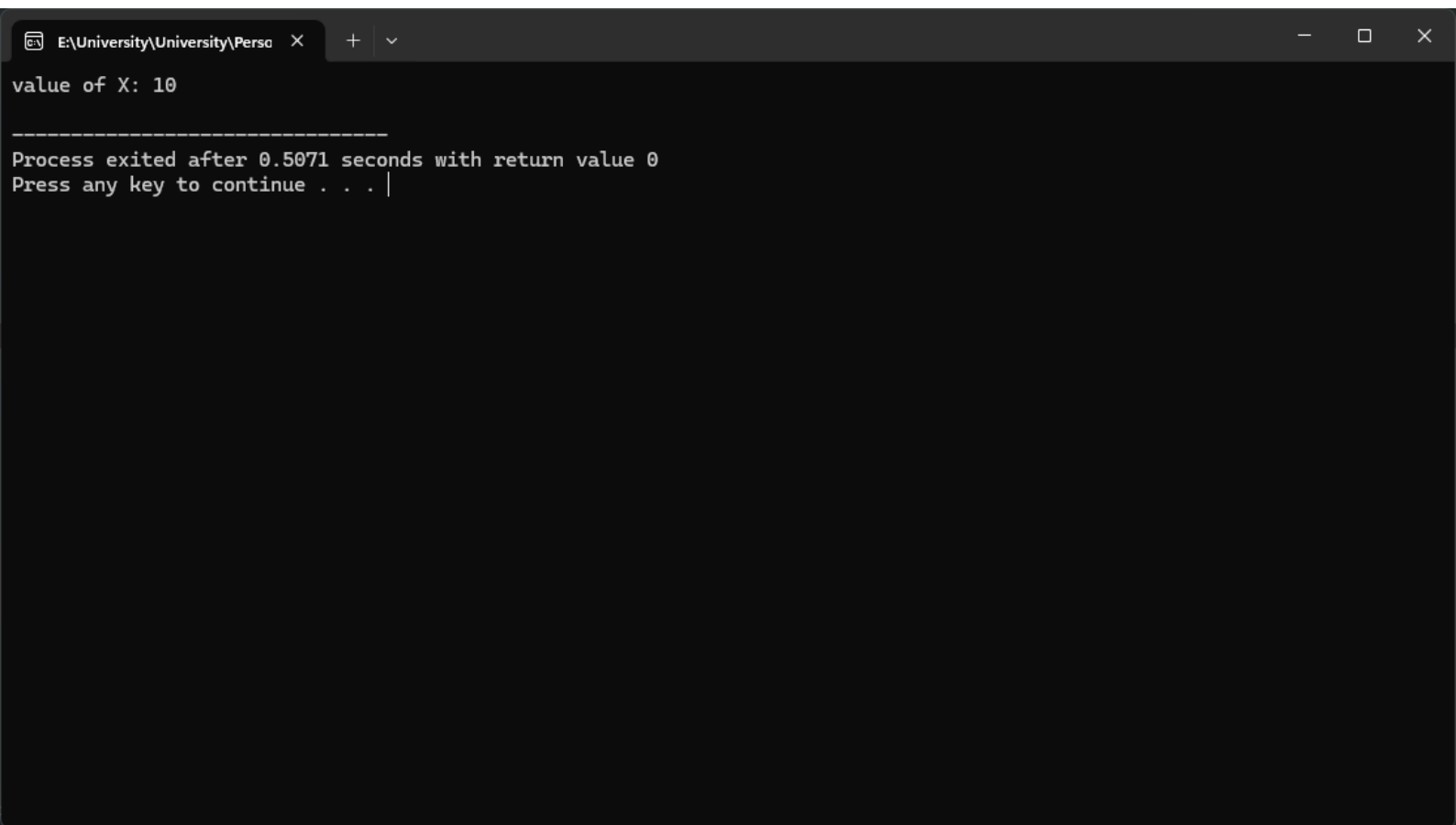
Example 5



A screenshot of a Windows command prompt window. The title bar shows the file path "E:\University\University\Perso" and standard window controls. The command prompt displays the following text:

```
Constant Function is called  
Non Constant Function is called  
20  
30  
-----  
Process exited after 0.1797 seconds with return value 0  
Press any key to continue . . . |
```

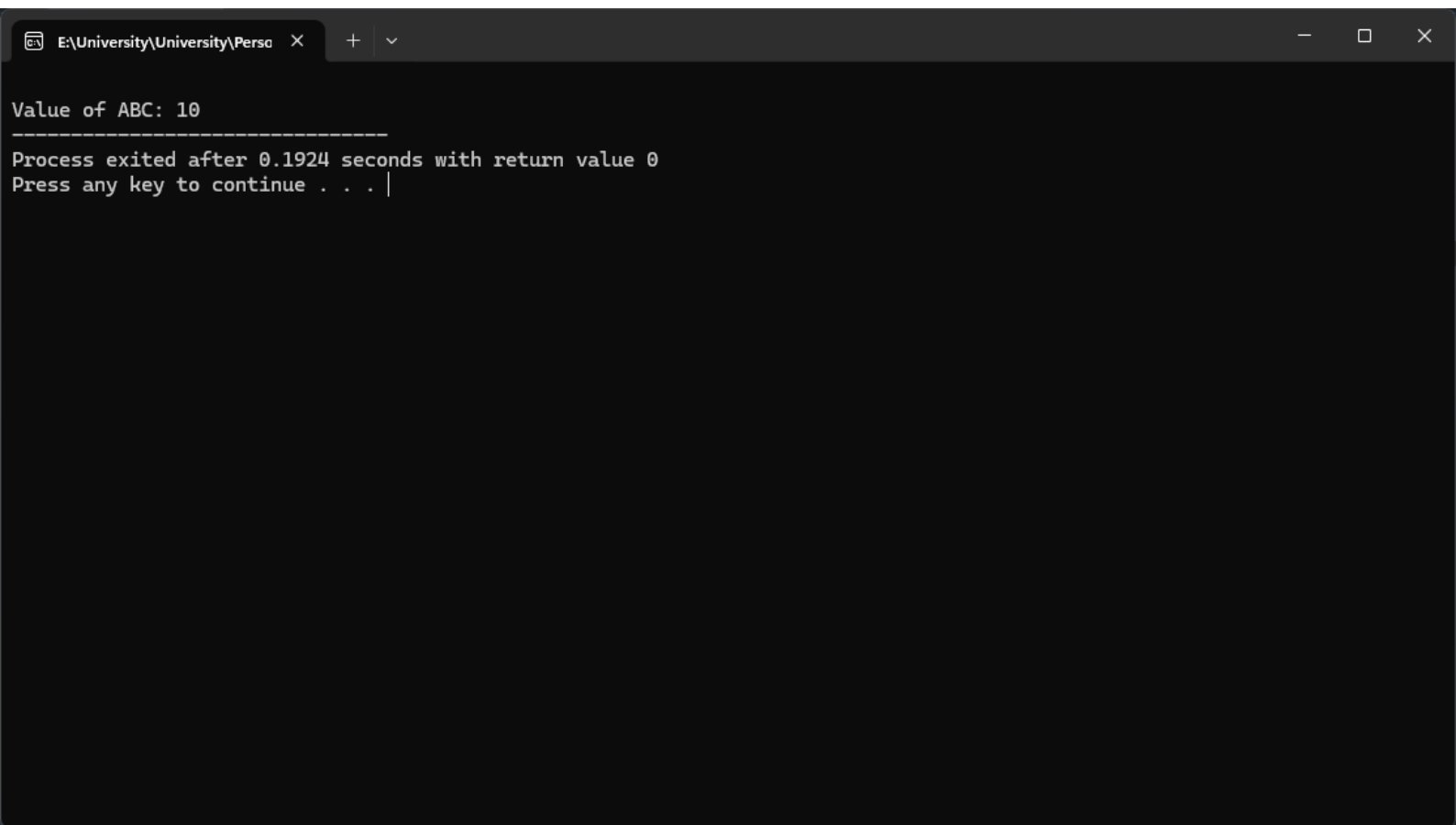
Example 6



A screenshot of a Windows command prompt window. The title bar shows the file path "E:\University\University\Perso" and standard window controls. The command prompt displays the output of a program execution. It shows "value of X: 10" followed by a separator line. Below the separator, it states "Process exited after 0.5071 seconds with return value 0" and "Press any key to continue . . . |".

```
E:\University\University\Perso >
value of X: 10
-----
Process exited after 0.5071 seconds with return value 0
Press any key to continue . . . |
```

Example 7



The image shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'E:\University\University\Perso' and includes standard window controls (minimize, maximize, close). The command prompt displays the following text:

```
Value of ABC: 10
-----
Process exited after 0.1924 seconds with return value 0
Press any key to continue . . . |
```